D2NN Loss Function Sweep

Static D2NN Practical Limits & Optimal Loss Strategy

2026-04-03 | FSO Beam Cleanup Research

Experiment Overview

10 loss strategies across 3 sweeps, 2 datasets

Sweep	Data	Strategies	Status
0401	dn100um (Lanczos)	PIB, Strehl, IO, CO+PIB	Complete
0402	pitch_rescale	Abs Bucket (F), TP+PIB (B), Multiplane (C)	Complete
0403	pitch_rescale	CO+hardTP, AbsBkt+CO, PIB+hardTP	In progress

10
Loss Strategies

5000
Training Pairs

30
Epochs Each

~6 min
Per Epoch (A100)

Normalized PIB Lies

High PIB does NOT mean high received power

Strategy	PIB@10μm	Throughput	Abs Power vs Turb	Verdict
PIB Only (0401)	**90.1%**	50.7%	**58.7%**	**LOSS**
Abs Bucket F (0402)	81.4%	**98.6%**	**105.8%**	**GAIN**
TP+PIB B (0402)	81.6%	**98.4%**	**105.8%**	**GAIN**
Multiplane C (0402)	84.2%	39.5%	**44.2%**	**LOSS**
CO+hardTP (0403)	74.0%	**99.3%**	97.0%	~NEUTRAL

Key Insight: PIB Only achieved 90% PIB but absolute received power was only 58.7% of turbulent baseline. **Throughput is the real metric.**

Why Does Throughput Drop?

1. **Phase Mask Pattern**
   D2NN learns high-frequency phase gratings to redirect energy

2. **Evanescent Waves**
   High spatial frequencies exceed propagation cutoff ($k^2x + k^2y > k^2$)

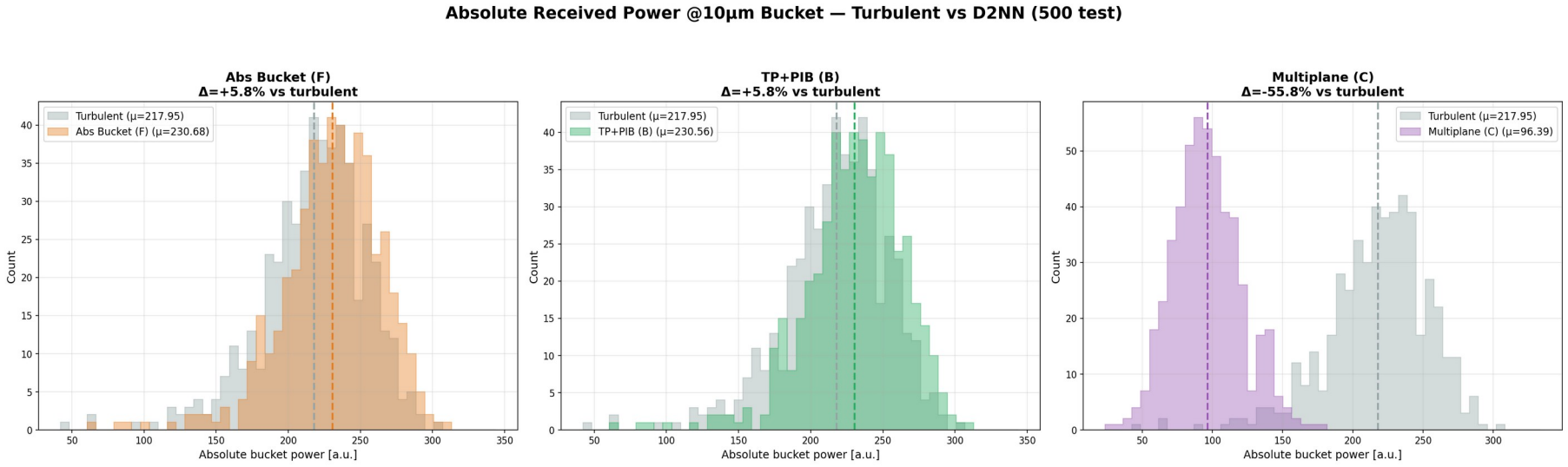
3. **Angular Spectrum Filter**
   Band-limited propagation zeros out evanescent components

4. **Energy Lost**
   Turbulent beams lose MORE energy (50%) than vacuum (20-30%)

D2NN acts as a spatial filter — scatters unwanted energy away

Absolute Received Power (0402)

Turbulent (gray) vs D2NN (color) — 500 test samples, bucket @10µm



Abs Bucket (F)
+5.8% vs turbulent

TP+PIB (B)
+5.8% vs turbulent

Multiplane (C)
-55.8% vs turbulent

Throughput vs PIB: Structural Trade-off

PIB@10μm vs Throughput

Strategy	PIB@10μm	Throughput	Abs Power
PIB Only	**90.1%**	**50.7%**	58.7%
Abs Bucket (F)	81.4%	**98.6%**	**105.8%**
TP+PIB (B)	81.6%	**98.4%**	**105.8%**
Multiplane (C)	84.2%	**39.5%**	44.2%
CO+hardTP	74.0%	**99.3%**	97.0%

Enforce TP

Scatter blocked → PIB limited to ~81% (+5.8% absolute power)

Release TP

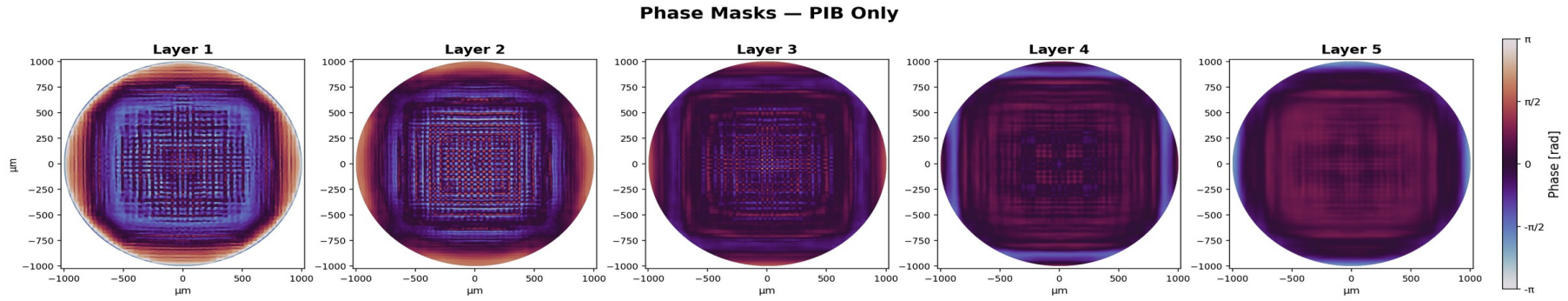
PIB reaches 90% but absolute power drops to 59% of baseline

Fundamental Limit

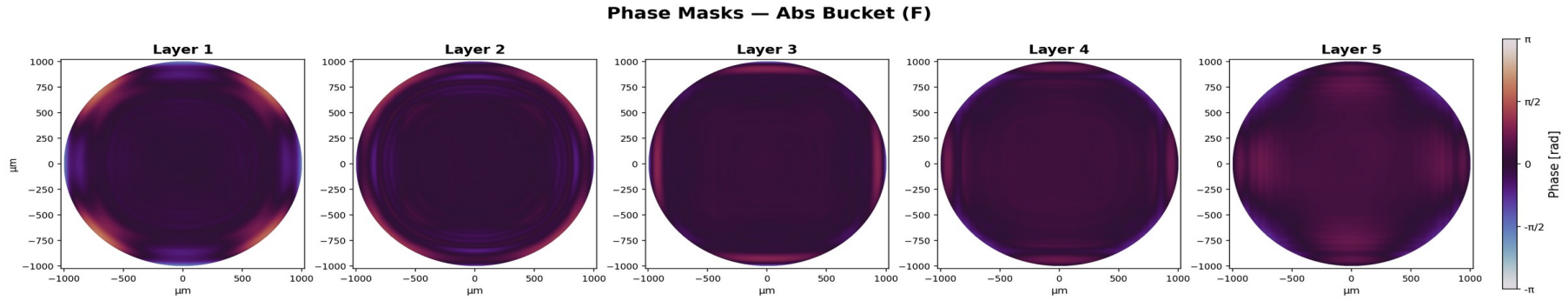
Static phase masks cannot correct random turbulence without energy loss

Phase Mask Patterns

PIB Only (0401) — High-frequency grating, aggressive scatter



Abs Bucket (0402) — Nearly flat, throughput preserved



Zernike Mode Analysis

Can D2NN correct wavefront aberrations while maintaining throughput?

Strategy	Higher-order Δ	Tip/Tilt Δ	Throughput	Assessment
F/B (Abs Bucket)	±1 nm	±0.2 nm	**98%**	No WF correction
Multiplane C	**-28 nm**	+1.3 nm	40%	Good WF, bad TP
PIB Only	-28 nm	+16 nm	51%	WF + scatter

Dilemma:

Wavefront correction requires the D2NN to create complex phase patterns.

These patterns generate high spatial frequencies that get clipped by propagation.

Result: correcting wavefront = losing energy. A structural limit of static optics.

CO Loss Alone Fails (0403)

co_hard_tp: CO + Quadratic TP penalty (λ=10)

99.3%	74.0%	97.0%
Throughput	PIB@10μm	Abs Power vs Turb
Best of all strategies	Below baseline (76.1%)	3% loss in received power

Why: CO optimizes full-field matching (amplitude + phase), not bucket concentration. **No gradient signal to focus energy into the 10μm bucket.**

Remaining hope: pib_hard_tp combines PIB concentration + TP enforcement

Conclusions

1

Throughput is the real metric

Normalized PIB can be misleading. Only strategies with TP > 95% improved absolute received power.

2

Static D2NN limit: +5.8%

Best absolute power improvement over turbulent baseline (Abs Bucket / TP+PIB strategies).

3

PIB + TP are structurally opposed

Concentrating energy requires scattering → throughput loss. Static masks can't escape this trade-off.

4

Vacuum PIB (94%) is unreachable

With throughput maintained, maximum PIB is ~82%. Closing the gap requires fundamentally different approaches.

Next Steps

Pending (0403)

abs_bucket_plus_co
Focal abs power + CO wavefront matching

pib_hard_tp
PIB maximization + quadratic TP (λ=10)

Most promising: PIB Only's 90% + TP enforcement

Future Directions

Dynamic D2NN
Sample-adaptive phase masks

Hybrid AO + D2NN
AO for low-order, D2NN for high-order

More Layers / Wider Aperture
Increase degrees of freedom

TV Weight Sweep
Optimize smoothness vs correction

Breaking the static limit requires moving beyond fixed phase masks